Department of Computer Science & Engineering

Course Syllabus: CS101 - Data Structures & Algorithms

Academic Regulations & Policies (Page 1)

1. Attendance Requirement: Students must maintain a minimum of **75% attendance** in both lectures and practical sessions to be eligible for university examinations. Students with attendance between 65% and 74% may apply for condonation on medical grounds with valid documentation.

2. Grading System: Continuous Internal Assessment (CIA) accounts for 40% of final marks. End Semester Examination (ESE) accounts for 60% of final marks. Passing mark is 40% overall.

Course Modules (Page 2)

Unit 1: Linear Data Structures - Arrays, Linked Lists, Stacks, Queues, Infix/Postfix Conversions, Applications.

Unit 2: Trees & Hierarchical Structures - Binary Search Trees, AVL Trees, Heaps, Priority Queues, Trie Data Structure.

Unit 3: Graph Algorithms & Dynamic Programming - BFS, DFS, Dijkstra's Shortest Path, Minimum Spanning Trees (Kruskal & Prim), Dynamic Programming principles, Knapsack Problem, Longest Common Subsequence.

Unit 4: Sorting & Searching Technologies - QuickSort, MergeSort, HeapSort, External Sorting, Hash Tables, Collision Resolution Strategies.